

4017 counter IC

The CD4017 IC is a CMOS Decade counter and it is used in the applications of low-range counting. This IC will count from 0 to 10 and the circuit with an IC 4017 will save board space as well as the time necessary to design the circuit. This CMOS IC is frequently used in designing the 10 [LEDs](#)-based circuits to run the light for beginners. So it is one of the most flexible counters because it counts up to 10 & also it includes 10 separate outputs. This IC includes the counter as well as the decoder.

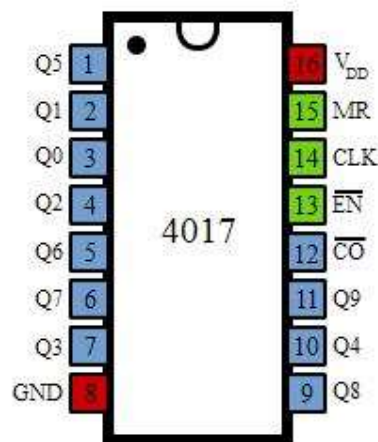
Pin Configuration of IC 4017

A CMOS decade counter like 4017 IC includes a five-stage Johnson counter & 10 decoded outputs for counting up to 10 decimals. This IC can be used in different counter circuits such as an LED chaser light & also a non-contact circuit with an AC detector. This IC includes 16-pins where 10 & 16 pins are output pins. This IC counts for every positive otherwise increasing edge input provided at the CLK input.

Here, the output begins from '0' and moves to output '9'. Once it reaches then the output will count at '9' and again it repeats from 0 & continues this revolution similar to a ring counter. On each count among 0 to 9, the particular output pin can include a high state & the remaining output maintains low; simply one o/p will be high at a time.

For instance, if the count is presently three then the output at pin-7 is will be in a high state whereas the remaining pins are in a less state. If two cycles of the square wave are applied toward the clock input, then the output will shift to output 4 on the positive edge of the primary cycle which alters output-4 to high state & output-3 to low state.

Now the condition remains until another positive edge of the next cycle arrives, after that the o/p moves to pin1 or output 5 and maintains this condition. The **ic 4017 internal structure** is shown below.



- Pin-1: It is the output 5. It goes high when the counter reads 5 counts.
- Pin-2: It is the output 1. It goes high when the counter reads 0 counts.

- Pin-3: It is the output 0. It goes high when the counter reads 0 counts.
- Pin-4: It is the output 2. It goes high when the counter reads 2 counts.
- Pin-5: It is the output 6. It goes high when the counter reads 6 counts.
- Pin-6: It is the output 7. It goes high when the counter reads 7 counts.
- Pin-7: It is the output 3. It goes high when the counter reads 3 counts.
- Pin-8: It is the Ground pin that should be connected to a LOW voltage (0V).
- Pin-9: It is the output 8. It goes high when the counter reads 8 counts.
- Pin-10: It is the output 4. It goes high when the counter reads 4 counts.
- Pin-11: It is the output 9. It goes high when the counter reads 9 counts.
- Pin-12: This is divided by 10 output which is used to cascade the IC with another counter to enable counting greater than the range supported by a single IC 4017.
- Pin-13: This pin is the disabled pin. In the normal mode of operation, this is connected to ground or logic LOW voltage. If this pin is connected to logic HIGH voltage, then the circuit will stop receiving pulses and so it will not advance the count irrespective of several pulses received from the clock.
- Pin-14: This pin is the clock input. This is the pin from where we need to give the input clock pulses to the IC to advance the count. The count advances on the rising edge of the clock.
- Pin-15: This is the reset pin that should be kept LOW for normal operation. If you need to reset the IC, then you can connect this pin to HIGH voltage.
- Pin-16: This is the power supply (Vcc) pin. This should be given a HIGH voltage of 3V to 15V for the IC to function.

So overall this IC will produce a counter depending on the pulses of the clock that we are sending CLK pin. For the clock we will use a 555 Timer IC to generate CLK pulses.

The internal component inside the IC is as follow:

Johnson counter:

is a type of sequential digital counter that generates a unique sequence of states in a cyclic manner.

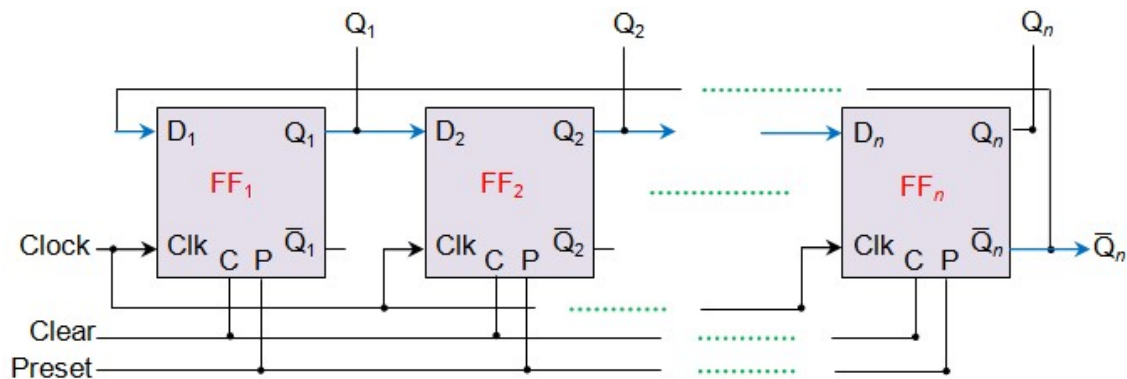


Figure 1 n -bit Johnson Counter Designed Using D Flip-Flops

Structure of a Johnson Counter

1. D Flip-Flops:

- The counter consists of a series of **D flip-flops** connected in a chain.
- The number of flip-flops (n) determines the total number of unique states ($2n$).

2. Feedback Mechanism:

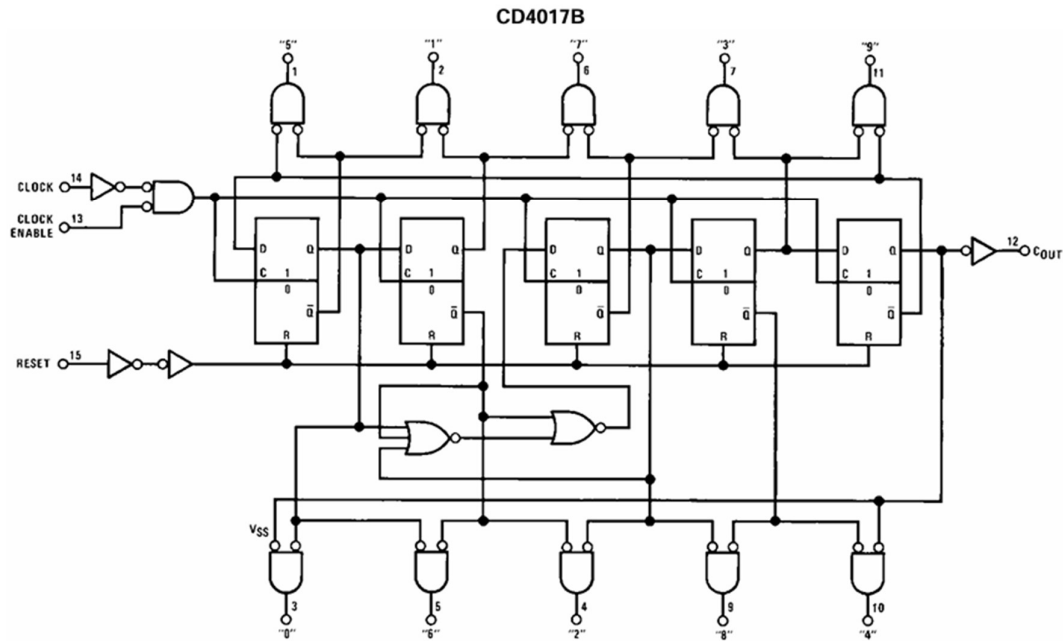
- The **inverted output** (Q -bar) of the last flip-flop is fed back into the input of the first flip-flop.
- This feedback is what distinguishes a Johnson counter from a simple ring counter.

This diagram shows the truth table of Johnson Counter circuit:

Clock Pulse No	FFA	FFB	FFC	FFD
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1

"1000", "1100", "1110", "1111", "0111", "0011", "0001", "0000"

Logic Diagrams



Terminal No. 8 = GND

Terminal No. 16 = V_{DD}

The internal structure of CD4017 uses Johnson Counter with a bit of configurations for enhancement using NAND gates to ensure efficiency and stability.

The NAND gate which are in the outputs of the counter makes sure to give the appropriate HIGH state depending in the pulses, so, the output will be low only when both inputs are HIGH, thus, if the output is low then it will face NOT gate which will let it to be HIGH.

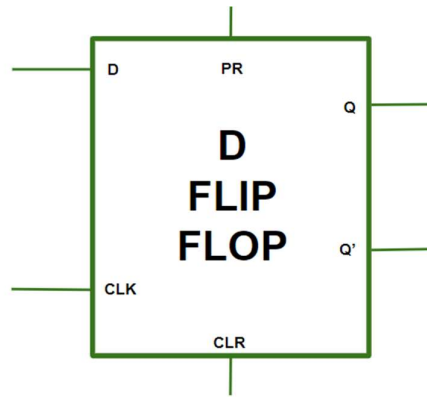
D-flip flop

The D Flip Flop Consists a single data input(D), a clock input(CLK),and two outputs: Q and Q' (the complement of Q).

Basic Structure

A D flip-flop consists of:

1. **Input (D):** The data to be stored.
2. **Clock (CLK):** Synchronizes when the data is read or written.
3. **Output (Q):** Stores and outputs the current data.
4. **Complement Output (Q-bar):** The inverted value of Q.



- Truth table

Clock Edge	D	Q _{next}	Q-bar
Rising/Falling	0	0	1
Rising/Falling	1	1	0
NO edge	x	Q _{previous}	Q-bar _{previous}

Operation of a D Flip-Flop:

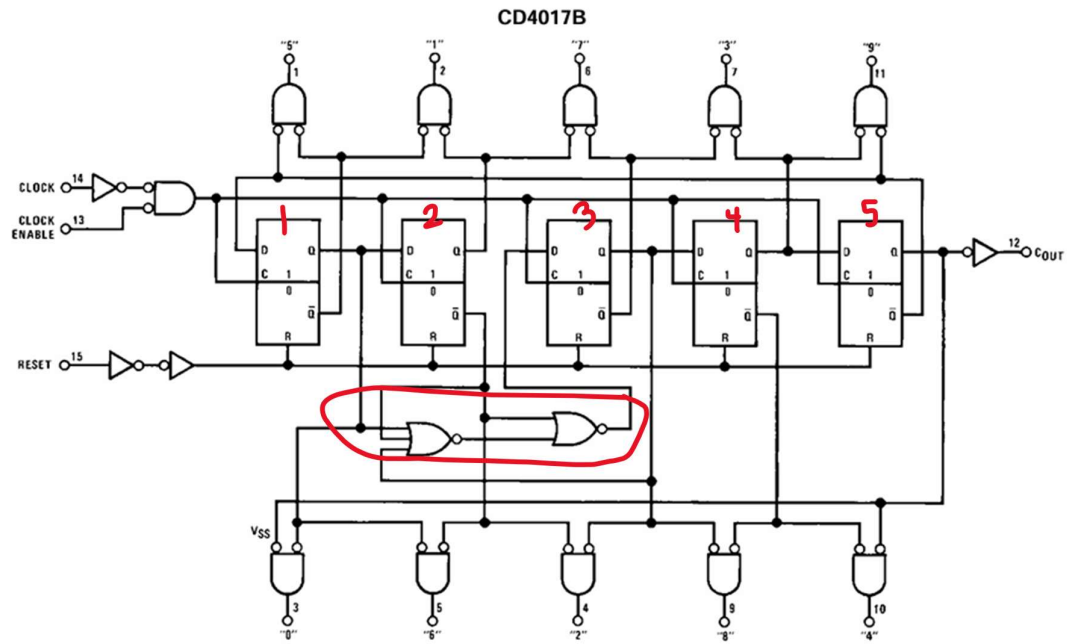
The D flip-flop is **edge-triggered**, which means it reacts only to the clock signal's edge (rising or falling edge). The key feature of the D flip-flop is that it **transfers the value of the D input to the Q output** only on the edge of the clock signal. Between clock edges, the output remains the same, regardless of changes in the D input.

1. Clock Edge (Rising or Falling):

- When the clock signal (CLK) receives a **rising edge** (from 0 to 1) or a **falling edge** (from 1 to 0), the **D input** is sampled, and the value of D is transferred to the **Q output**.
- The Q output will hold the value of the D input at that instant until the next clock edge.

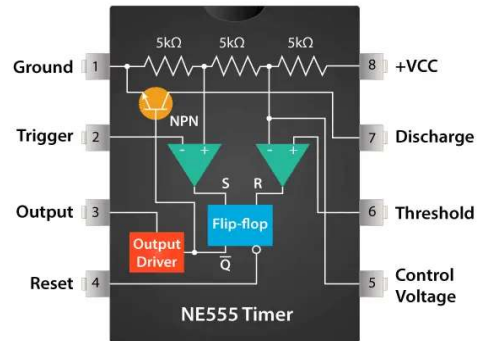
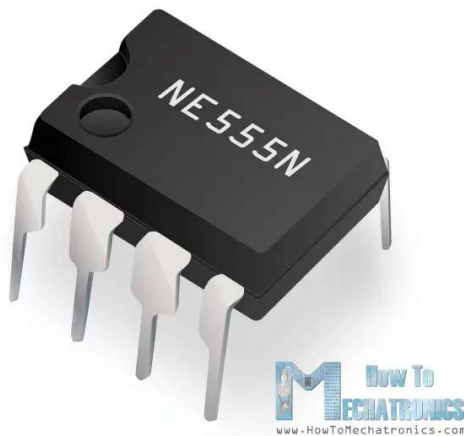
2. No Clock Edge:

- Between clock edges, no change happens to the output. The output Q remains the same, regardless of any changes in the D input.

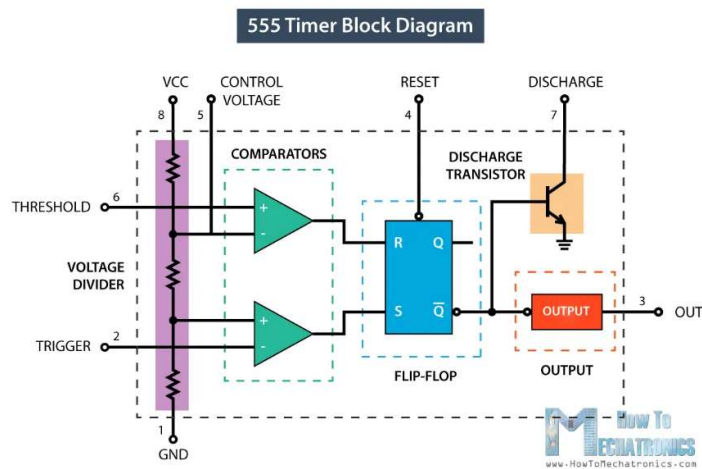


These OR gates insures that when the first, fourth flip-flop is in a low state and second in high state, third flip-flop data will have HIGH state.

555 Timer

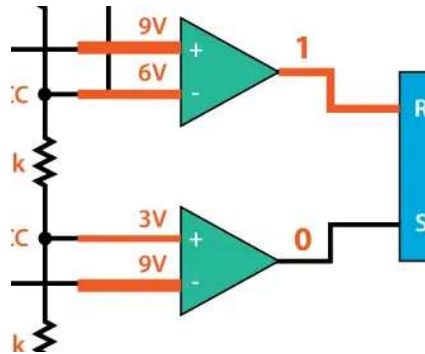


- How it works?

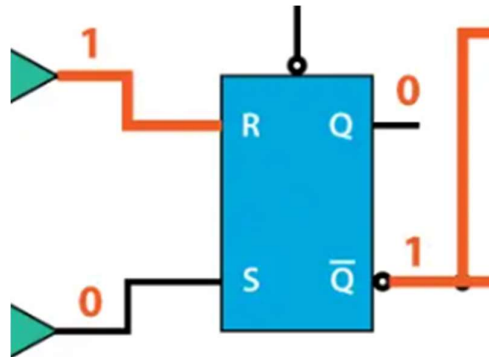


Represented with a block diagram it consists of 2 comparators, an RS flip-flop, a voltage divider, a discharge transistor and an output stage.

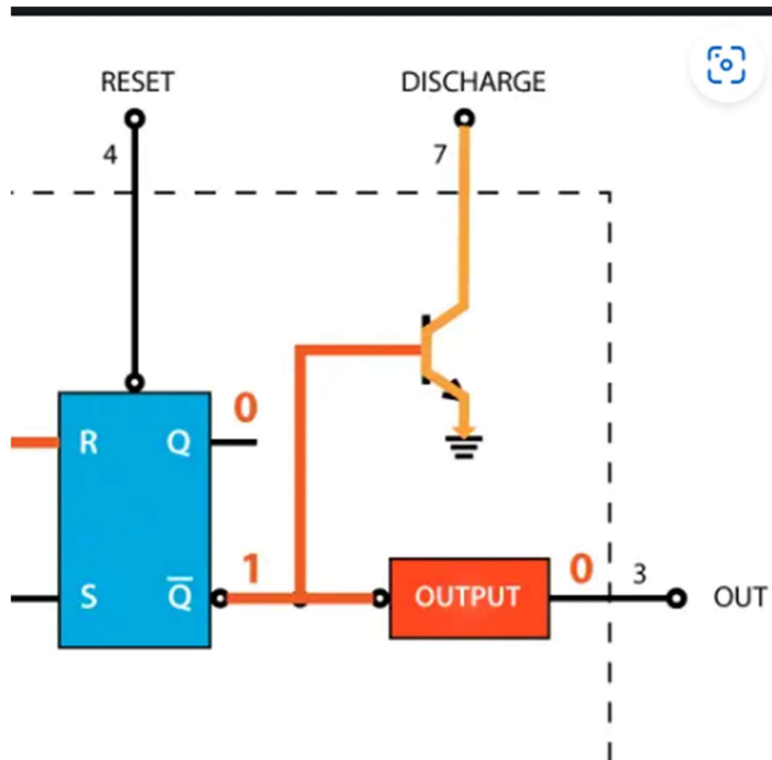
Voltage divider: The voltage divider consists of three identical 5k resistors which create two reference voltages at $1/3$ and $2/3$ of the supplied voltage, which can range from 5 to 15V.



Comparators: Next are the two comparators. A comparator is a circuit element that compares two input voltages at its **positive (non-inverting)** and **negative (inverting)** input terminal. If the input voltage at the positive terminal is higher than the input voltage at the negative terminal the comparator will output 1. Vice versa, if the voltage at the negative input terminal is higher than the voltage at the positive terminal, the comparator will output 0. This is illustrated in the next picture.



RS flip-flop: using the three pins, Trigger, Threshold and Control, we can control the output of the two comparators which are then fed to the R and S inputs of the flip-flop. The flip-flop will output 1 when R is 0 and S is 1, and vice versa, it will output 0 when R is 1 and S is 0. Additionally the flip-flop can be reset via the external pin called "Reset" which can override the two inputs, thus reset the entire timer at any time.



The Q-bar output of the flip-flop goes to the output stage or the output drivers which can either source or sink a current of 200mA to the load. The output of the flip-flop is also connected to a transistor that connects the “Discharge” pin to ground.

By referring to the datasheet of 555Timer we can choose the values of the resistors and capacitor from the giving equation, note that we are using Astable mode 555Timer configuration:

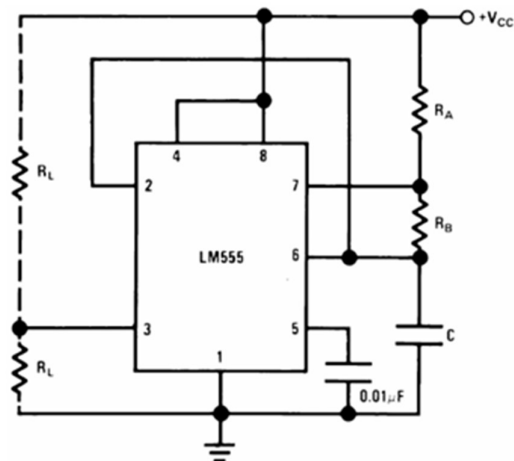


Figure 15. Astable Waveforms

The charge time (output high) is given by:

$$t_1 = 0.693 (R_A + R_B) C \quad (1)$$

And the discharge time (output low) by:

$$t_2 = 0.693 (R_B) C \quad (2)$$

Thus the total period is:

$$T = t_1 + t_2 = 0.693 (R_A + 2R_B) C \quad (3)$$

The frequency of oscillation is:

$$f = \frac{1}{T} = \frac{1.44}{(R_A + 2R_B) C} \quad (4)$$

Figure 16 may be used for quick determination of these RC values.

The duty cycle is:

$$D = \frac{R_B}{R_A + 2R_B} \quad (5)$$

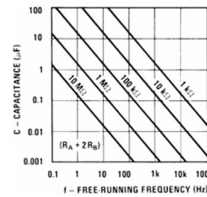
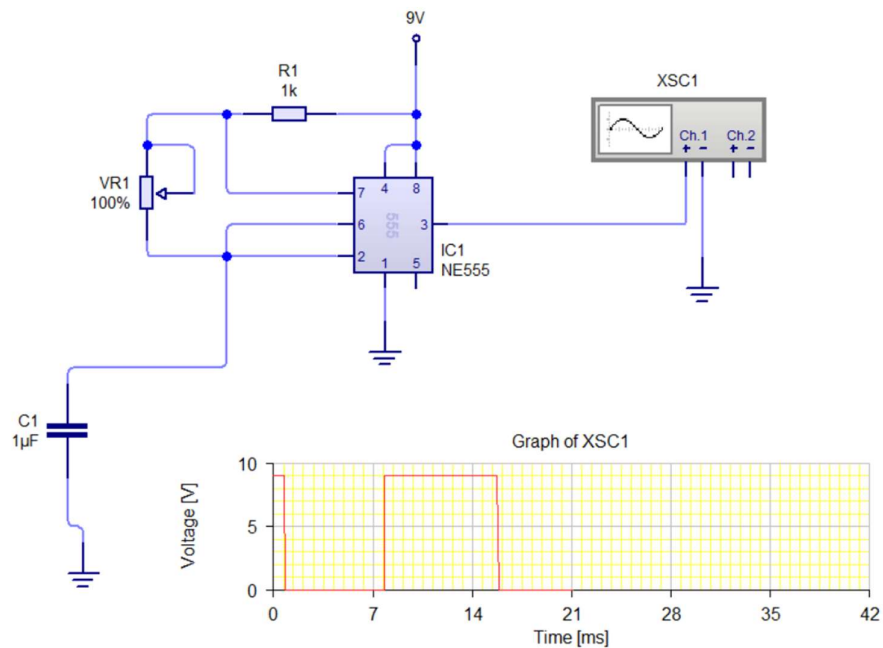
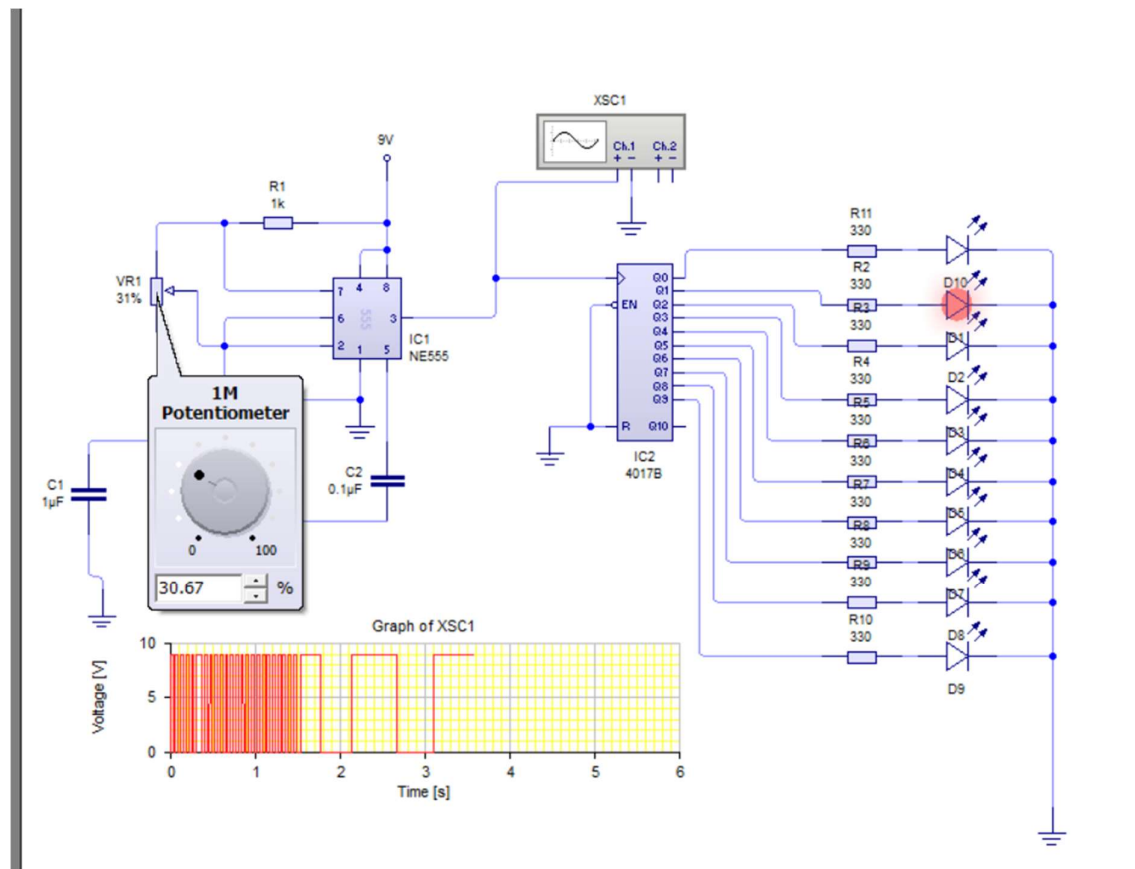


Figure 16. Free Running Frequency



The circuit above will help us to generate a PWM signals which we can control its frequency to input it to the CD4017 IC and get our sequential activation board.



This finalized circuit approves our explanation for both ICs and by varying the potentiometer value the frequency changes, thus the LEDs switching will correspond to the change.

**** I know the simulation was not part of the task but it was an essential step to proof what I wrote in this documentation.****